

ALI BAYOUMI

INDEPENDENT PORTFOLIO CASE STUDY

Planning & Project Controls

G+9 office building + roof

Primavera P6 / Baseline, update and management reporting

A complete building programme

1,206

scheduled activities

G+9 + roof

Engineering through handover

1,876

logic relationships

195

WBS nodes, including project root

33

resources

Engineering · Procurement · Construction · MEP · Finishes · External works · Handover

One reporting cut connects the model

Data date: 01 Apr 2027. Dates, cost and resources are reconciled before reporting.

01 / Establish

02 / Calculate

03 / Interpret

Baseline & status

Preserve the original baseline.
Update actuals and remaining work to the data date.

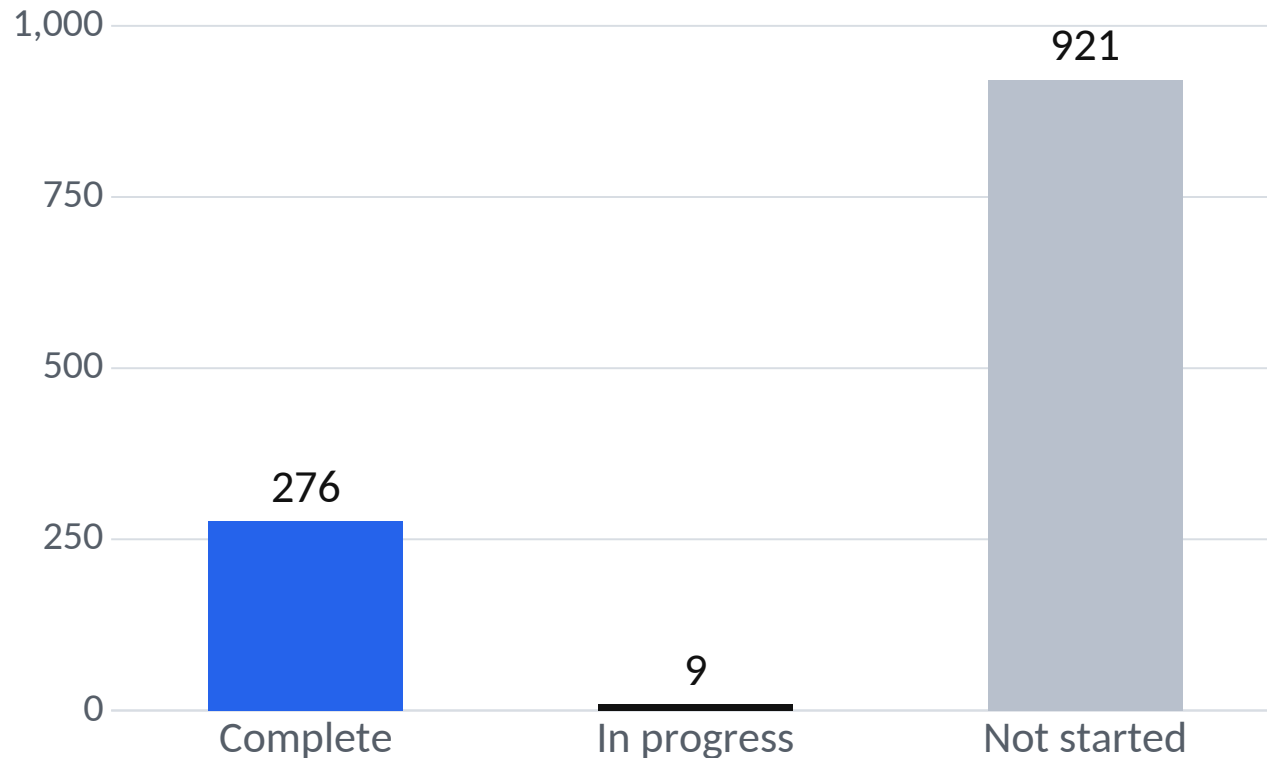
CPM & resources

Reschedule the logic.
Review resource leveling and preserved early / late dates.

Value & priorities

Reconcile earned value.
Issue the critical-path, cost, resource and lookahead reports.

276 activities complete; 930 remain open



01 APR 2027

A single status snapshot across the integrated programme.

9

activities in progress

Counts describe activity status. They are not a weighted physical-progress percentage.

Keep the three finish dates distinct

Final project handover: saved early dates and leveled dates answer different questions.

15 AUG 2028

16 AUG 2028

22 AUG 2028

CPM early finish

Leveled handover

Baseline handover

Preserved scheduled early date in the source snapshot.

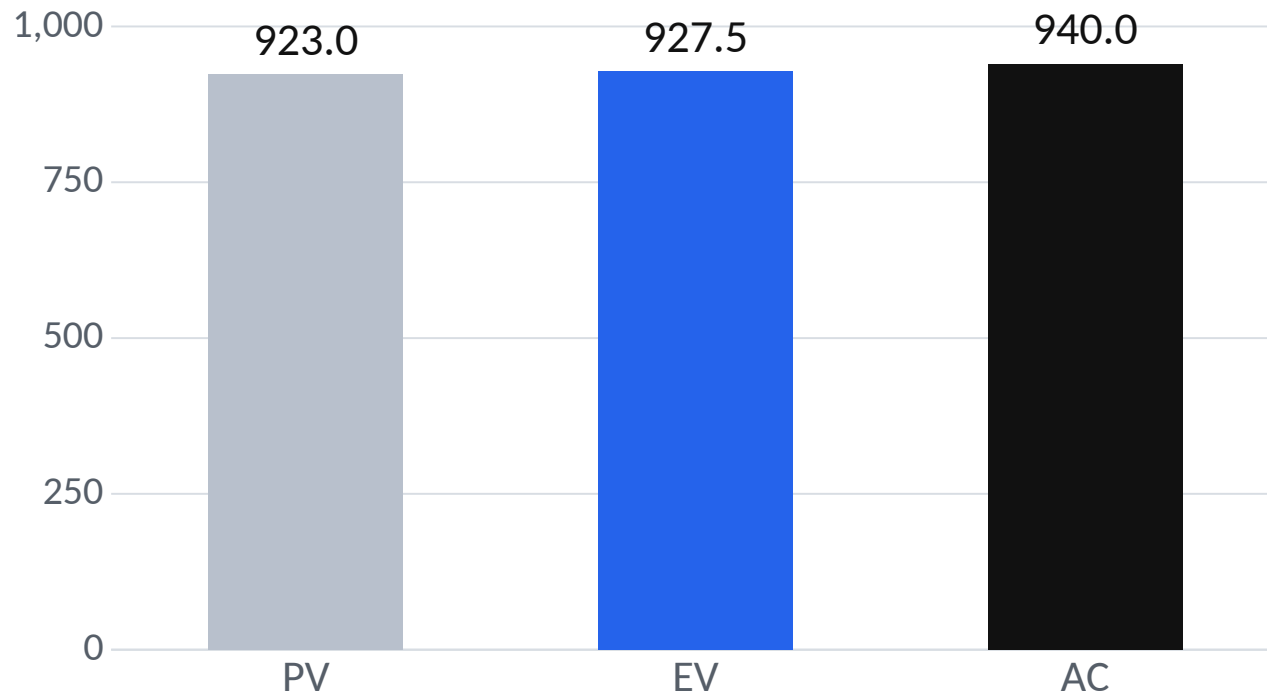
Current HO-023 remaining / leveled finish.

Original baseline HO-023 finish. Use the same milestone scope.

Saved comparison: leveled handover is 6 calendar days before baseline. This is not a tested recovery claim.

The EVM totals reconcile to the source

Cumulative value at 01 Apr 2027 · USD thousands



PV \$923,042.33 · EV \$927,476.92 · AC \$940,042.33

1.0048

SPI = EV / PV

0.9866

CPI = EV / AC

The SPI gain is entirely reporting LOE

+\$4,434.59

Schedule variance

Produced by PM-038
Monthly Progress Reporting

Value in PM-038	USD
Planned value	96,452.33
Earned value	100,886.92

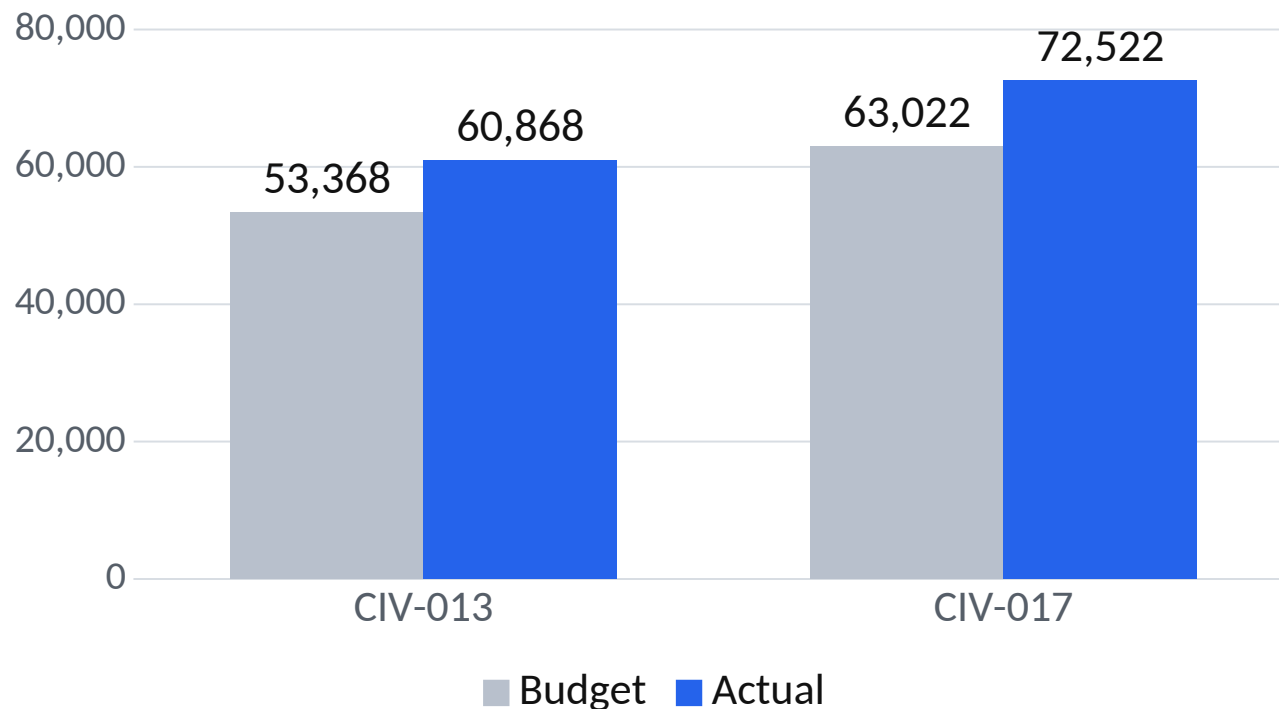
Outside this LOE

PV = EV = \$826,590.00

Positive SPI alone does not establish construction progress ahead of plan.

Two completed activities explain the overrun

Completed foundation work · USD



\$17,000

overrun retained in EAC

CIV-013 +\$7,500

CIV-017 +\$9,500

EAC \$5,268,281

BAC \$5,251,281

Configured forecast: $EAC = \text{actual cost} + \text{remaining cost}$. Both completed activities have zero ETC.

Protect the pour, then the full handover

Ground-floor milestones need both the current finish and the correct baseline reference.

09 APR 2027

25 APR 2027

05 MAY 2027

Column curing

Beam / slab pour

Structural handover

GF-ST-007

192 hours remaining

Total float: 9 hours

GF-ST-015

Baseline: 19 Apr 2027

6 calendar days later

GF-ST-019

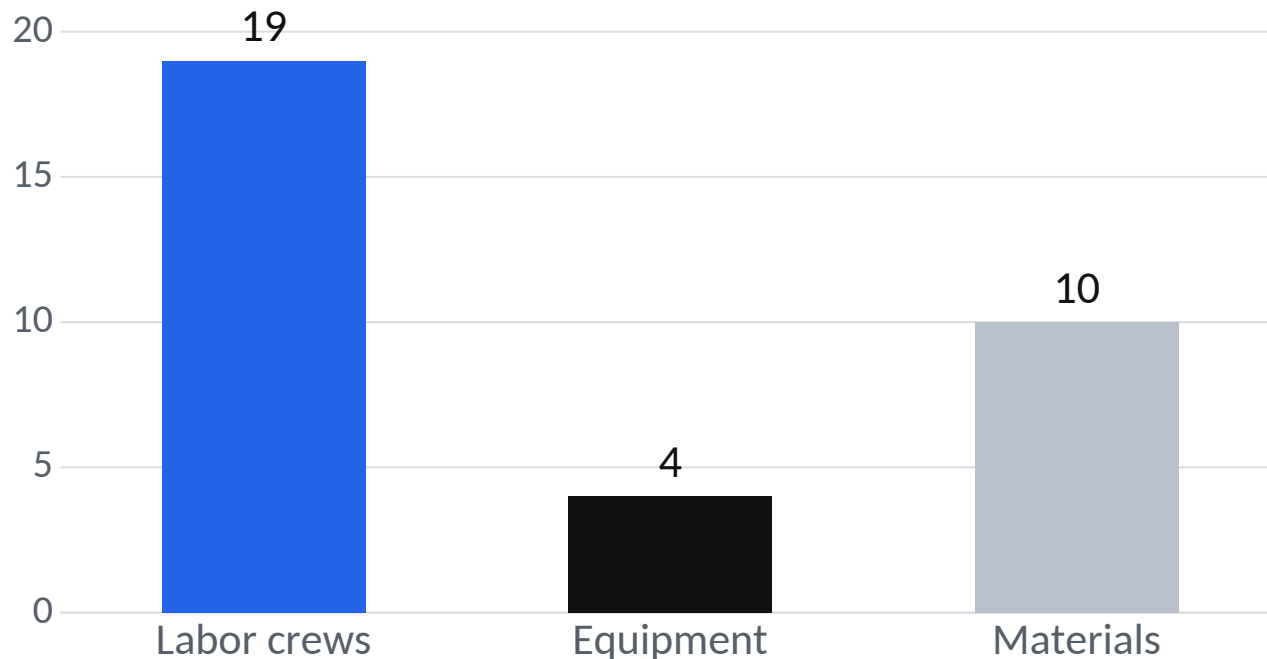
Baseline: 01 May 2027

4 calendar days later

Formwork → reinforcement → sleeves → inspection → pour. Validate curing and floor-release assumptions.

Leveling needs more than a finish date

33 resources · 572 assignments



31

direct resource delays in the log

222

incomplete activities with different
saved early and remaining dates

These are different measures. The log also records 469 predecessor delays and zero unable to level.

The next 90 days define the workfronts

01 APR → 30 JUN 2027

189 rows

Procurement finish gates in the saved lookahead

Package	Manufacture	FAT	Delivery	Approval
HVAC / Electrical / Plumbing / Low current	14 Apr	19 Apr	03 May	09 May
Fire fighting	21 Apr	26 Apr	10 May	16 May

Coordinate GF structure, masonry, MEP first-fix readiness and repetitive floor production.

The audit makes model limitations visible

Diagnostic	Source result	What it means
Network integrity	0 cycles / broken references	A connected graph still needs sound handoffs.
Incomplete activity float	184 zero; 496 high float	Review finish control and interfaces.
Relationship logic	112 positive lags; 12 SS-only exits	Confirm lag purpose and finish handoffs.
GF report baseline	7 of 8 BL1 rows differ	Use XML baseline 4516 for comparison.
Calendars / cost coverage	Open assumptions	Validate exceptions and package mapping.

High float is screened above 44 own-calendar working days. These are diagnostics, not a DCMA compliance score.

Three actions drive the next update

01

Protect structure

Confirm curing, stripping and next-floor release assumptions. Track the pour and full GF handover separately.

02

Release MEP gates

Track manufacture, FAT, delivery, inspection and approval. Keep June 30 starts inside the lookahead.

03

Control exceptions

Explain completed-work overruns. Reconcile leveling delays and correct the report baseline assignment.

The reporting pack stays linked to the model

Original P6 output

OFF-04 · Earned Value Summary · 01 Apr 2027

Planned Value Cost	Eamed Value Cost	Actual Total Cost
\$923,042.33	\$927,476.92	\$940,042.33

The source amounts underpin the editable EVM charts and the variance analysis.

Five P6 reports

Labor crew loading
Earned value summary
PV / EV / AC curve
GF structural extract
90-day lookahead

Original outputs are retained alongside the corrected analytical comparisons.

A model that can be explained and defended

Build the logic. Reconcile the story. Define the next action.

SOURCE MODEL

Current-project XER
XML with the original baseline

CONTROL OUTPUTS

Five P6 reports
Corrected analytical extracts

TECHNICAL DEFENSE

Audit and reconciliation
Documented assumptions

Contractor Programme Review

C-01 Calendar exceptions require confirmation

Review item	Observation and requested response
Source finding	The saved programme has no holiday exceptions for 2026–2028.
Potential effect	Work could fall on dates when crews are unavailable. The time impact is unquantified.
Review comment	Submit the agreed holiday and shutdown calendar. Explain any continuous-work calendars.
Required action	Planner: update a copy, recalculate CPM and compare the same handover milestone.

Training response due: 02 Apr 2027. Status: open pending evidence.

Independent training exercise using assumed events and data; not a live contractual claim.

Progress Verification & Reporting

Slab package: 400 m³ total scope. Reporting cut: 01 Apr 2027.

Reporting basis	Quantity	Progress	Training evidence
Planned to date	200 m³	50%	Assumed quantity plan
Contractor reported	180 m³	45%	Assumed daily report
Verified in exercise	160 m³	40%	Mock Q-001 + IR-001

-10 percentage points

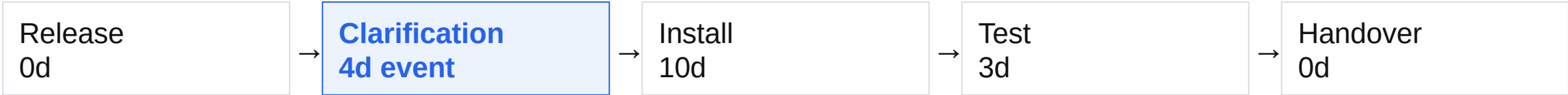
Verified vs planned. Quantity shortfall: 40 m³.

Log	Finding	Action owner / response	Due
D-01	40 m³ below plan	Site manager: submit recovery plan	02 Apr 2027
V-01	20 m³ unverified	Site engineer: reconcile records	02 Apr 2027

Independent training exercise using assumed events and data; not a live contractual claim.

Delay Analysis: Illustrative TIA

Model data date: 05 Apr 2027, 08:00. Mon–Fri, 8h/day, no holidays.



Inserted fragnet: clarification before installation. All links FS, zero lag.

Remaining work	Before event	After event
Installation, 10 working days	05–16 Apr 2027	09–22 Apr 2027
Testing, 3 working days	19–21 Apr 2027	23–27 Apr 2027
Training handover, M-TR01	21 Apr 2027, 17:00	27 Apr 2027, 17:00

Impact: +4 working days / +6 calendar days

EOT review also needs the contract, event responsibility and supporting records.

FIDIC procedure learning: confirm form, edition and Particular Conditions.

Independent training exercise using assumed events and data; not a live contractual claim.